

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			B1	ALL BOLTS ARE METRIC DIA.
20										B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
20	U04		MODULAR DBB	K06AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		
30	U04		MODULAR DBB	K06AC06	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
40	U04	VAV	MODULAR DBB	K06AC06	3	3	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
50	U04		MODULAR DBB	M06AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
60	U04		MODULAR DBB	M06AC06	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
70	U04	VAW	MODULAR DBB	M06EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2		
80	U04	VAW	MODULAR DBB	M06EC02	1	1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3		
90											
90		VB2					***** BALL VALVES *****				
100											
100	U04		BALL	B06AC01	0.75	1.5	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
110	U04		BALL	B06AC05	2	4	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB LO	0	0		
120	U04		BALL	B06AC07	6	6	BALL 600# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
130											
130							***** RISING STEM BALL VALVES *****				
140											
140		VA5	BALL	R06AC06	2	4	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0		
150											
150							***** GATE VALVES *****				
160											
160	U04		GATE	G06AC48	0.75	1.5	GATE 600# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW	0	0		
170			GATE	G06AC41	2	2	GATE 600# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0		
180			GATE	W06AC70	3	4	GATE 600# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
190											
190							***** PIPE ABOVE GROUND *****				
200											
200			PIPE	5358508	0.75	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0		
210			PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
220			PIPE	5358505	3	3	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0		
230											
230	U04		PIPE	9006210	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE	0	0		
240			PIPE	9001255	8	8	LINEPIPE SMLS WT= 7.11 MM API-5L X60 BE	0	0		
250											
250							***** PIPE BELOW GROUND *****				
260											
260	U04		PIPE	9006210CTD	4	4	LINEPIPE SMLS WT= 6.02 MM API-5L X60 BE COATED	0	0		
270											
270							***** FITTINGS ABOVE GROUND *****				
280											
280			ELBOW	5392600	0.75	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
290			ELBOW	5392438	3	3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0		
300			ELBOW	5392606	0.75	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
310			ELBOW	5392444	3	3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0		
320											
320			TEE	5392609	0.75	2	TEE BW SCH XS LTCS A420 WPL6	0	0		
330			TEE	5392447	3	3	TEE BW SCH STD LTCS A420 WPL6	0	0		
340			TEE RED	5392661	0.75	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1	2		
350			TEE RED	5392454	0.75	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3		
360											
360			CAP	5392610	0.75	2	CAP BW SCH XS LTCS A420 WPL6	0	0		
370			CAP	5392448	3	3	CAP BW SCH STD LTCS A420 WPL6	0	0		
380	U04		CAP	9006215	4	4	CAP BW WT 6.02 MM LTCS A860 WPHY-60	0	0		
390											
390			RED CONC	5392458	0.75	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	1	2		
400			RED ECC	5392461	0.75	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1	2		
410			RED CONC	5392452	1.5	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	3		
420			RED ECC	5392453	1.5	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	3		
430	U04		RED ECC	9006216	2	2	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4		
440	U04		RED ECC	9006216	2	2	REDUCER ECC WT 6.02 MM / XS WT LTCS A860 WPHY-60	4	4		
450	U04		RED ECC	9006217	3	3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4	4		
460	U04		RED ECC	9006217	3	3	REDUCER ECC WT 6.02 MM / STD WT LTCS A860 WPHY-60	4	4		
470			RED ECC	9001373	4	4	REDUCER ECC WT 7.11 MM / 6.02 MM LTCS A860 WPHY-60	8	8		
480											

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 67.5 BARG. @ 75 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 75 °C	12/16/2024	BY OZAC	U04	Page 1 of 2	600H5M

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480							*****FITTINGS BELOW GROUND*****			
490										
490							***** INDUCTION BENDS ABOVE GROUND*****			
500										
500	U04		INDUC BEND	9006212	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
510	U04		INDUC BEND	9006213	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
520	U04		INDUC BEND	9006214	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0	
530										
530							***** INDUCTION BENDS BELOW GROUND*****			
540										
540	U04		INDUC BEND	9006212CTD	4	4	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
550	U04		INDUC BEND	9006213CTD	4	4	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
560	U04		INDUC BEND	9006214CTD	4	4	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.02 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	
570										
570							***** OLETS ABOVE & BELOW GROUND*****			
580										
580	U04		FLANGE	5504007	0.75	2	NIPOFLANGE 600# RF SCH STD/XS LTCS A350 LF2 CL.1	3	3	
590	U04		FLANGE	9006220	0.75	2	NIPOFLANGE 600# RF 6.02 MM/XS WT LTCS A694 F60	4	4	
600										
600	U04			9301695	1	1	WELDOLET WT 6.02MM/SCH XS LTCS A694 F60	4	4	
610										
610							***** FLANGES ABOVE GROUND*****			
620										
620	U04		FLANGE	5557036	0.5	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
630	U04		FLANGE	5557033	3	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
640			FLANGE	9006218	4	4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM	0	0	
650			FLANGE	9006218	4	4	FLG WN 600# RF LTCS A694 F60 WT 6.02MM	0	0	
660	U04		BLIND	9006222	0.75	4	FLG BLIND 600# RF LTCS A694 F60	0	0	
670										
670							***** FLANGES BELOW GROUND*****			
680							***** GASKETS *****			
690										
690			GASKET	5673103	0.5	4	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
700										
700		B1					***** BOLTING *****			
710										
710	U04		BOLTING	5675770	0.5	4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
720										
720			SPEC BLIND	5749587	0.75	4	SPECTACLE BLIND 600# RF LTCS A516-GR70	0	0	
730										

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